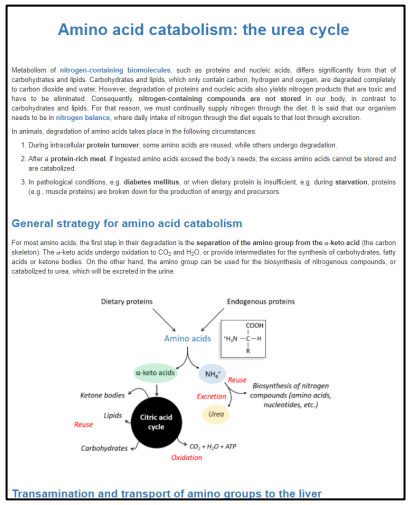
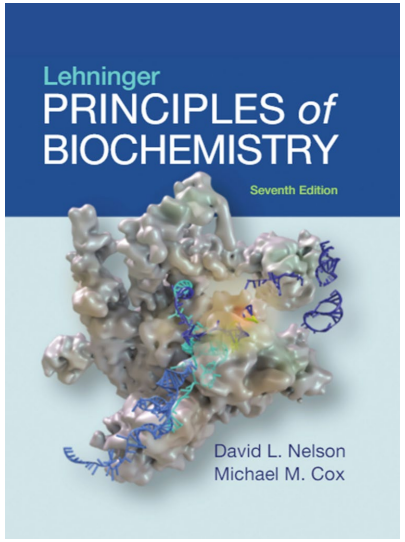


Amino acid catabolism

The urea cycle

Supporting material





Additional study material

ADI document:

Amino acid catabolism: the urea cycle

Reading

Lehninger's PoB Chapter 18:

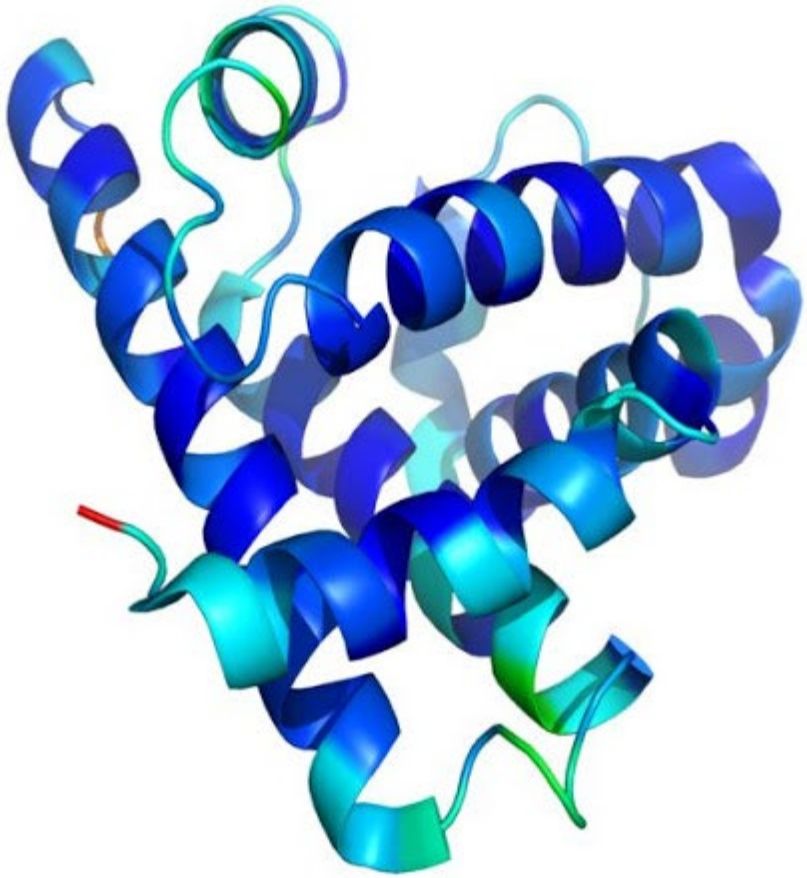
18.1. Metabolic fates of amino acids

18.2. The urea cycle

Learning goals

- General strategy for amino acid degradation
- Transamination
- Transport of amino nitrogen to the liver
- General characteristics of the urea cycle
- End products of amino acid degradation
- Main cofactors involved in amino acid metabolism
- Overview of genetic disorders affecting amino acid catabolism

Metabolism of nitrogen-containing biomolecules



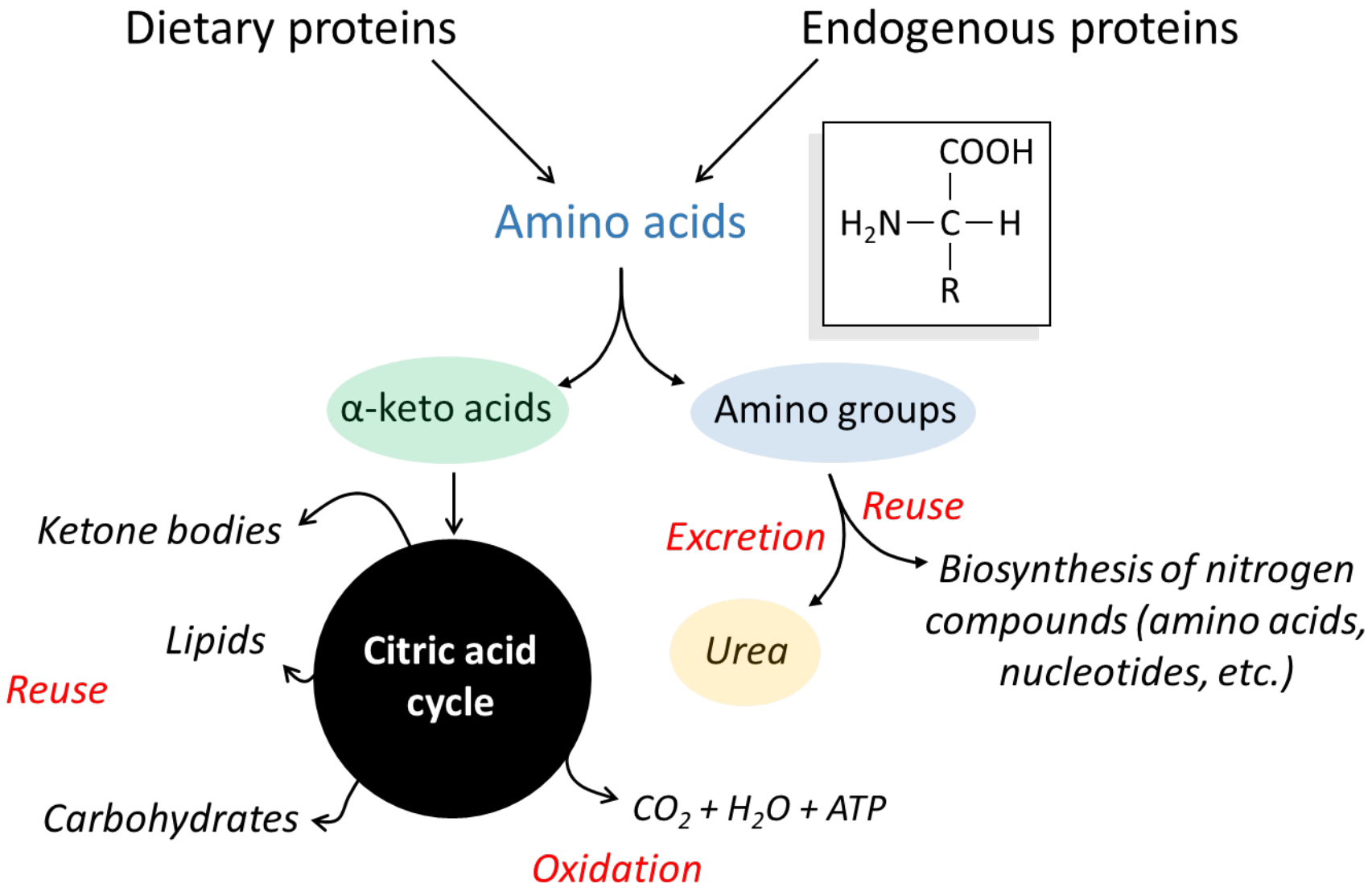
C H O N

Amino acids are not stored in a specialized reserve form

Circumstances of amino acid degradation

- Normal protein turnover
- Diet
- Starvation, diabetes mellitus

Outline of amino acid degradation



Transamination

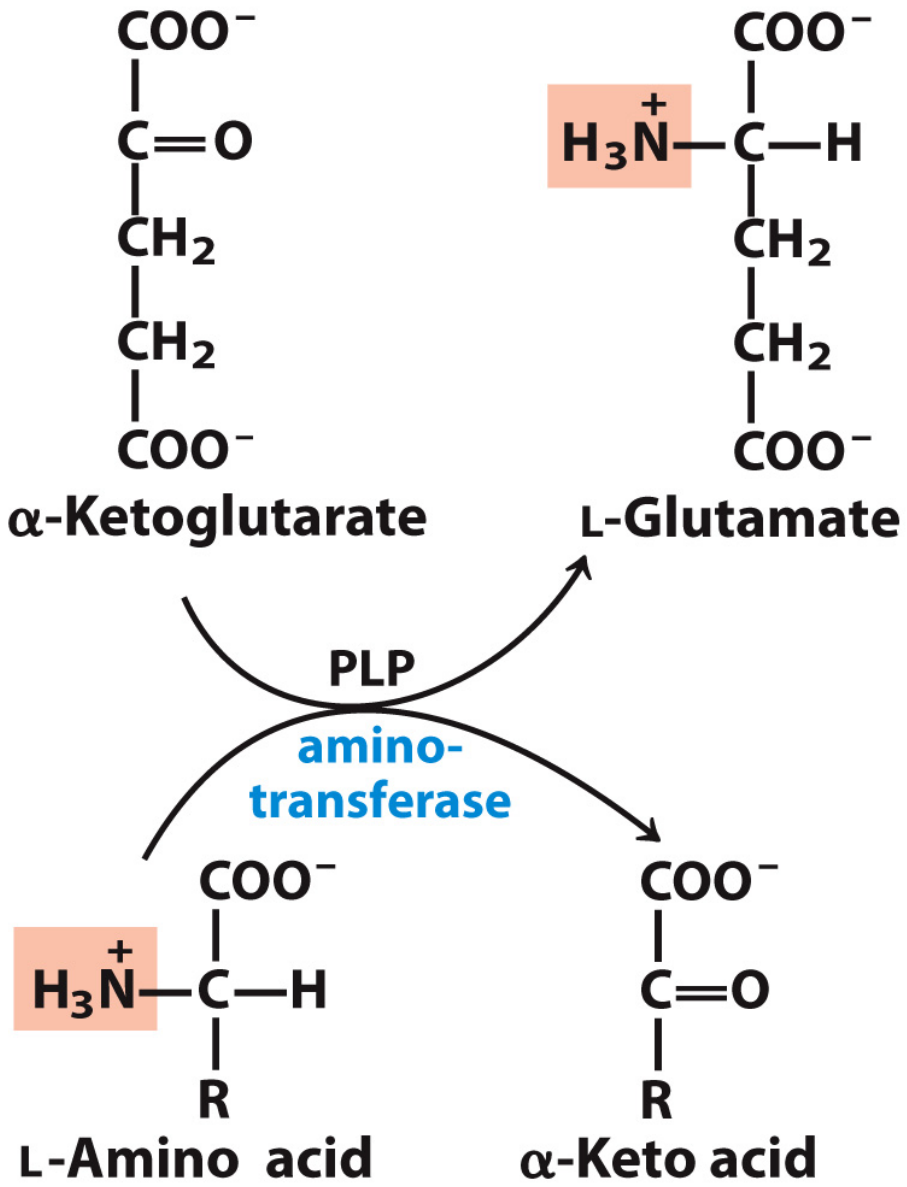
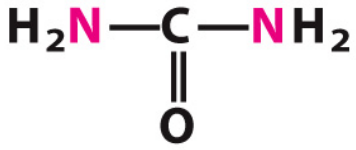


Figure 18-4
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Ways to excrete nitrogen



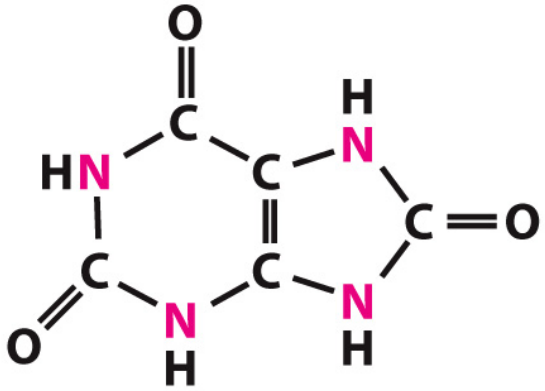
Ammonia (as ammonium ion)



Urea

Ammonotelic animals: most aquatic vertebrates, such as bony fishes and the larvae of amphibia

Ureotelic animals: many terrestrial vertebrates; also sharks

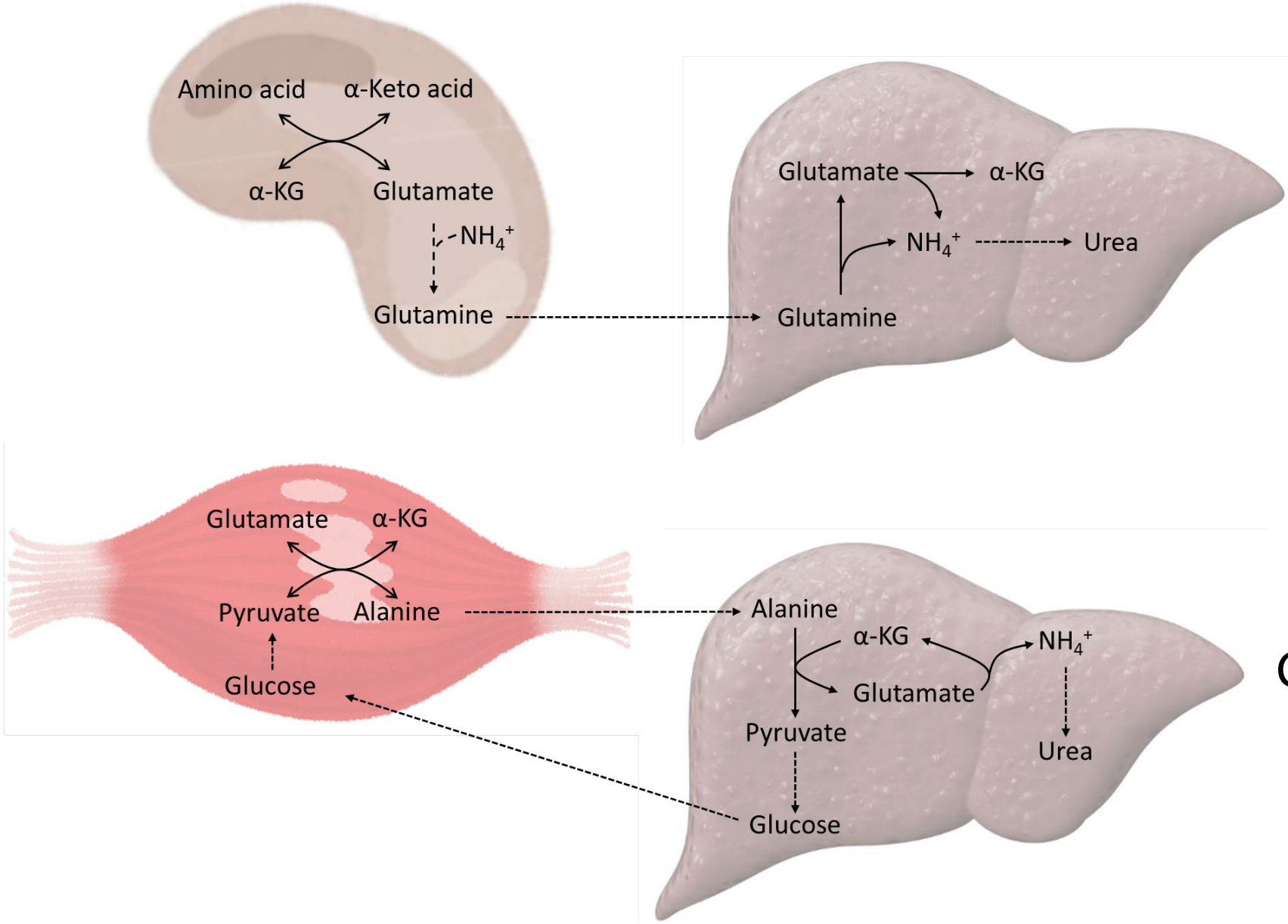


Uric acid

Uricotelic animals: birds, reptiles

Figure 18-2b
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Transport of amino nitrogen to the liver

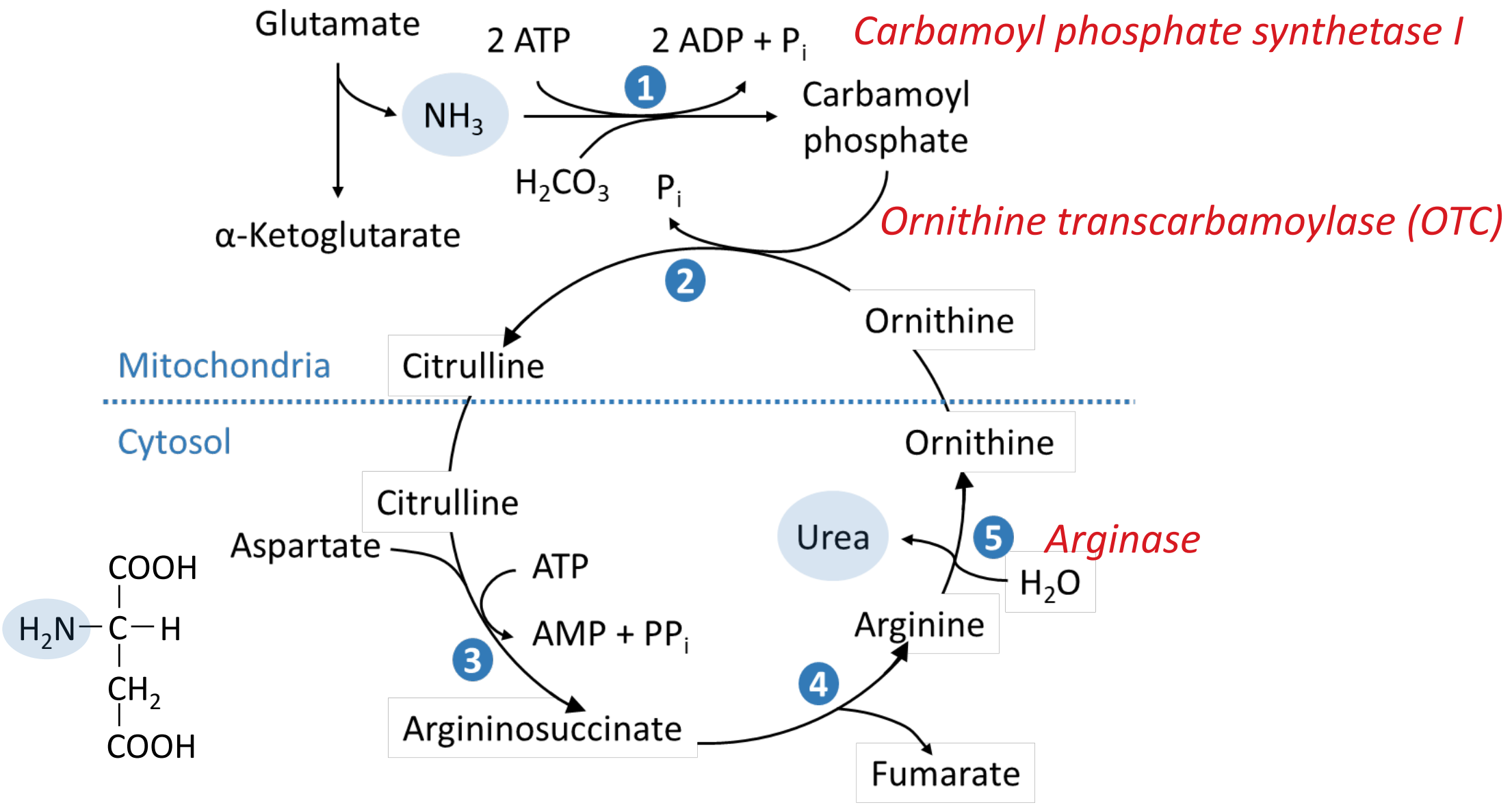


Glucose-alanine cycle

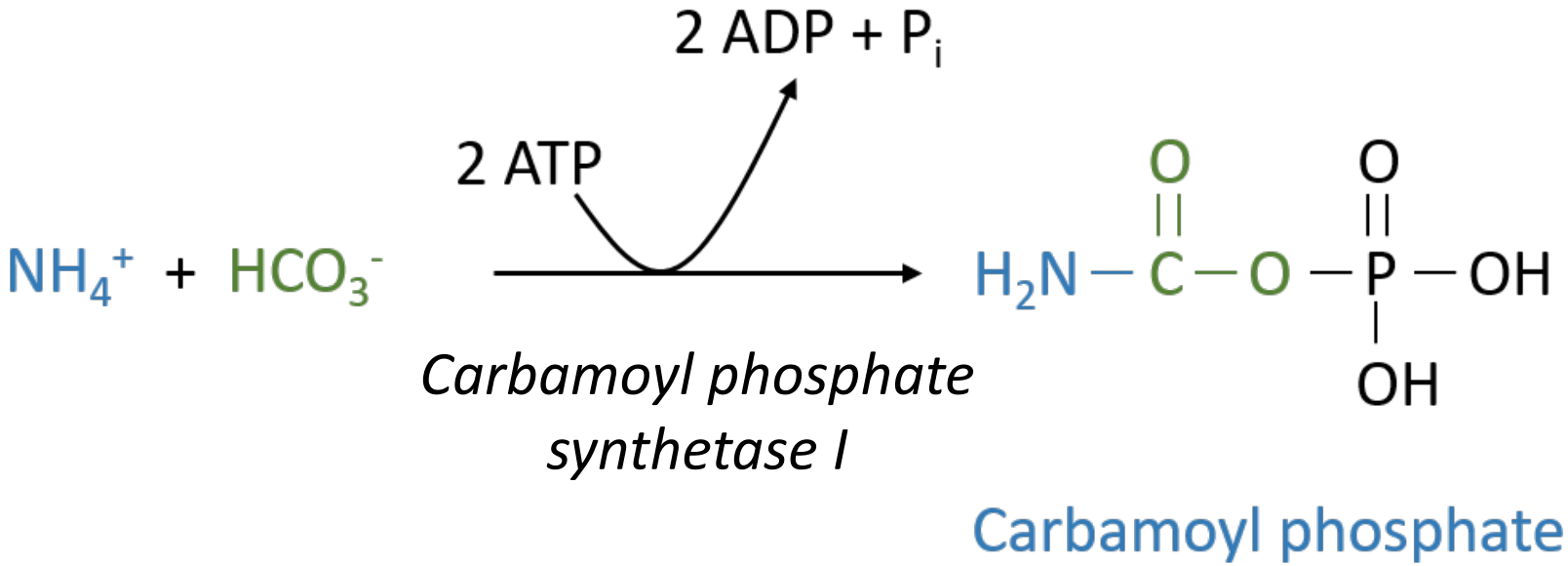
The urea cycle

- Described by Hans Krebs and Kurt Henseleit in 1932
- Nitrogen enters the urea cycle as NH_4^+ and aspartate and is converted to urea for excretion
- The urea cycle takes place almost exclusively in the liver
- The urea cycle spans two cellular compartments: mitochondria and cytosol
- The urea cycle is composed of five steps
- Once formed, urea passes into the bloodstream and travels to the kidneys, where it is excreted in the urine

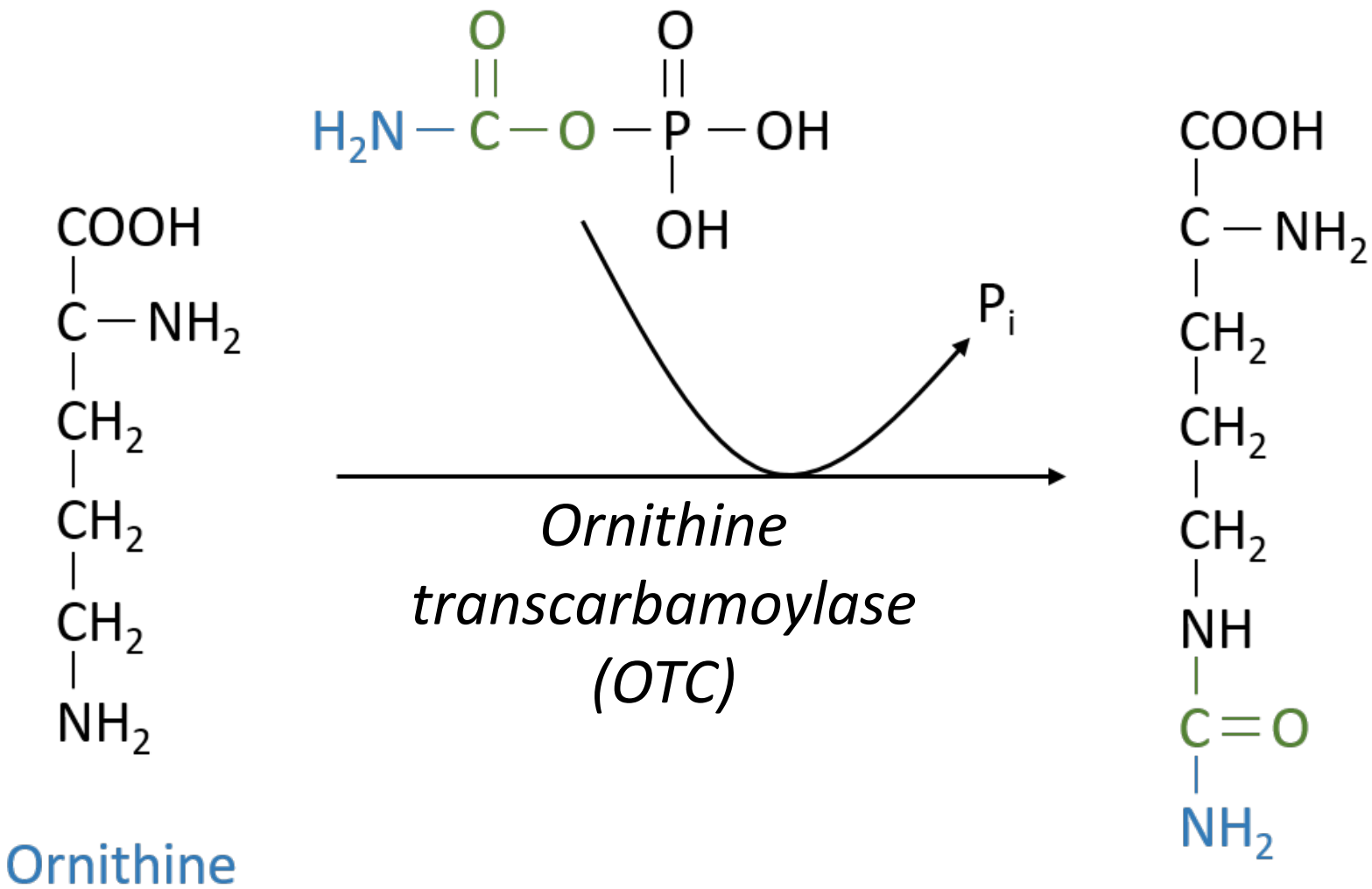
The urea cycle



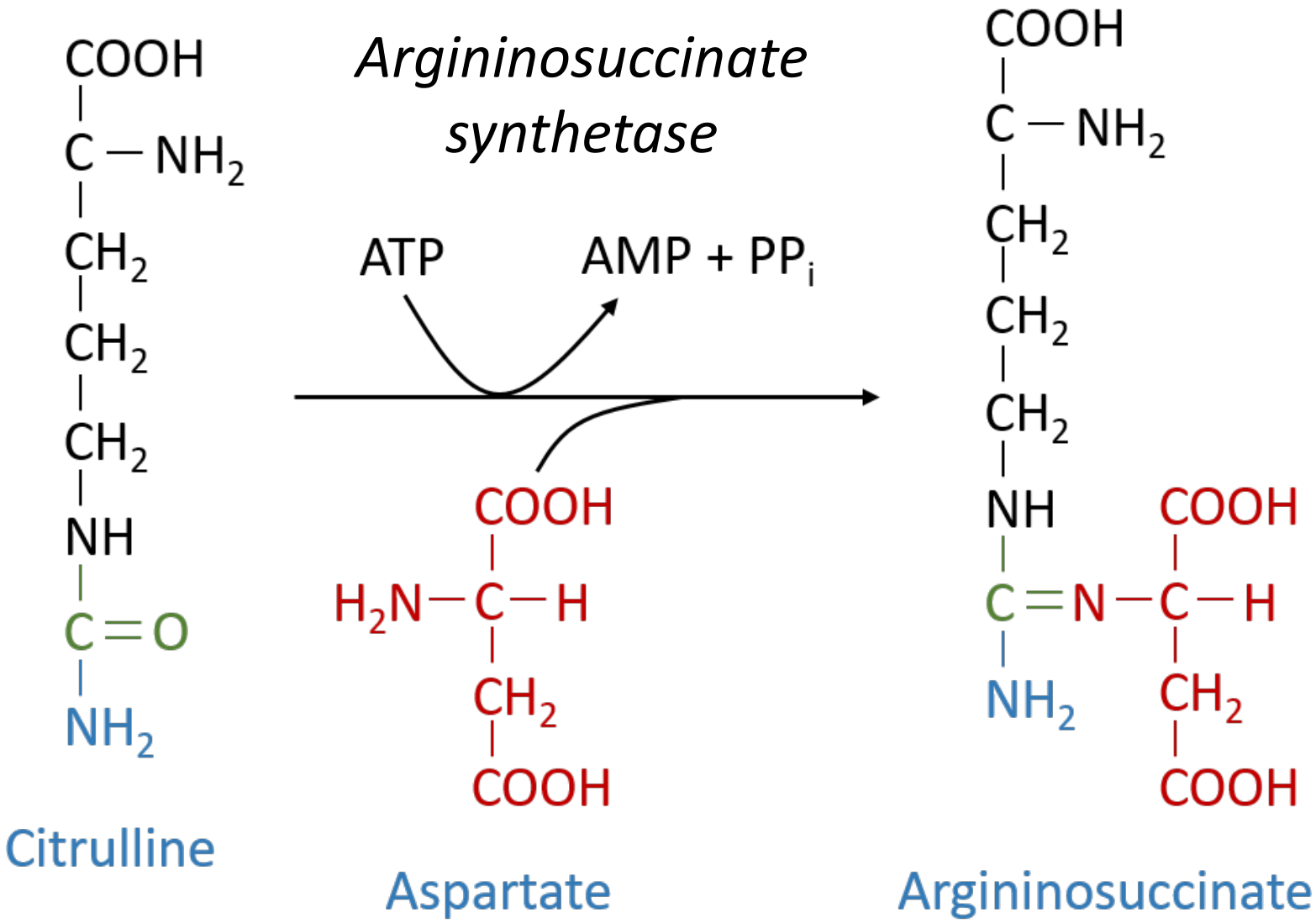
Incorporation of ammonium into carbamoyl phosphate



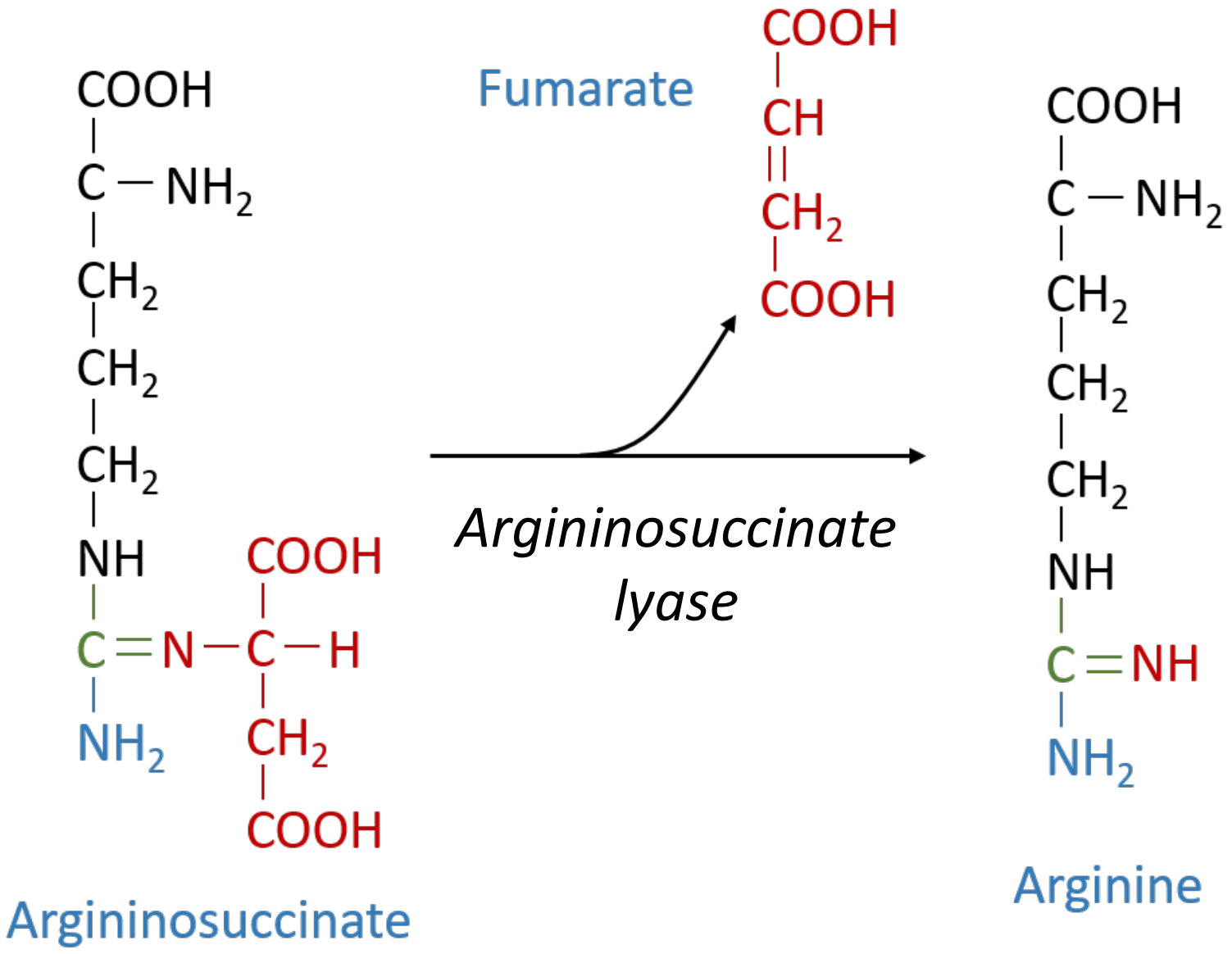
Citrulline synthesis



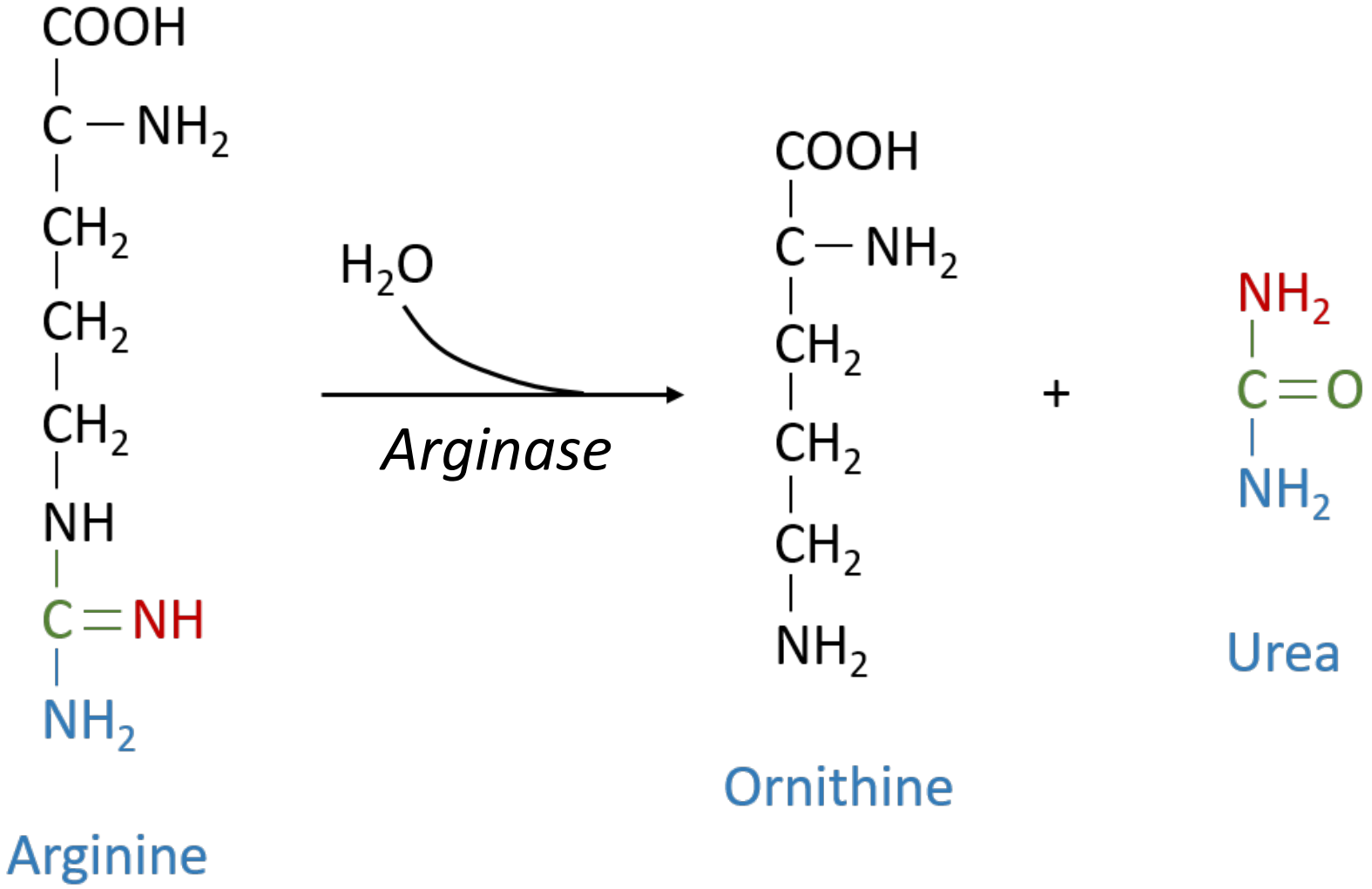
Argininosuccinate synthesis



Arginine synthesis



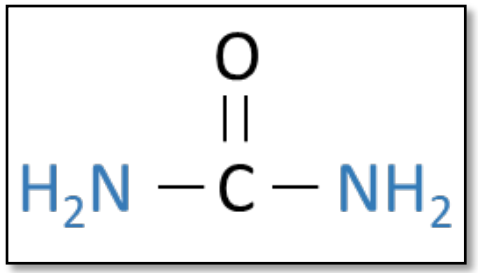
Urea formation



Overall balance of the urea cycle

$CO_2 + NH_3 + 3\ ATP + aspartate + 2\ H_2O$

$\rightarrow urea + 2\ ADP + AMP + 4\ P_i + fumarate$



Regulation of the urea cycle

- Long-term:

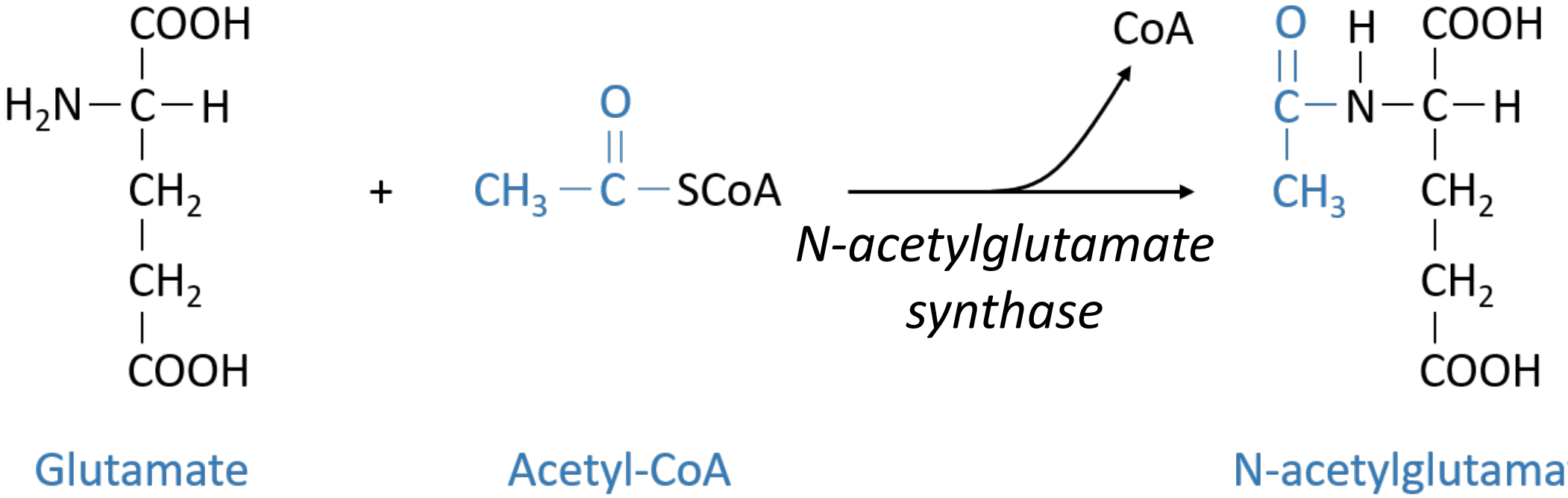
  Expression of urea cycle enzymes increases
  - High protein diet
  - Starvation

- Short-term:

  Arginine activates *N-acetylglutamate synthase*

  N-acetylglutamate activates *carbamoyl phosphate synthetase I*

Regulation of the urea cycle



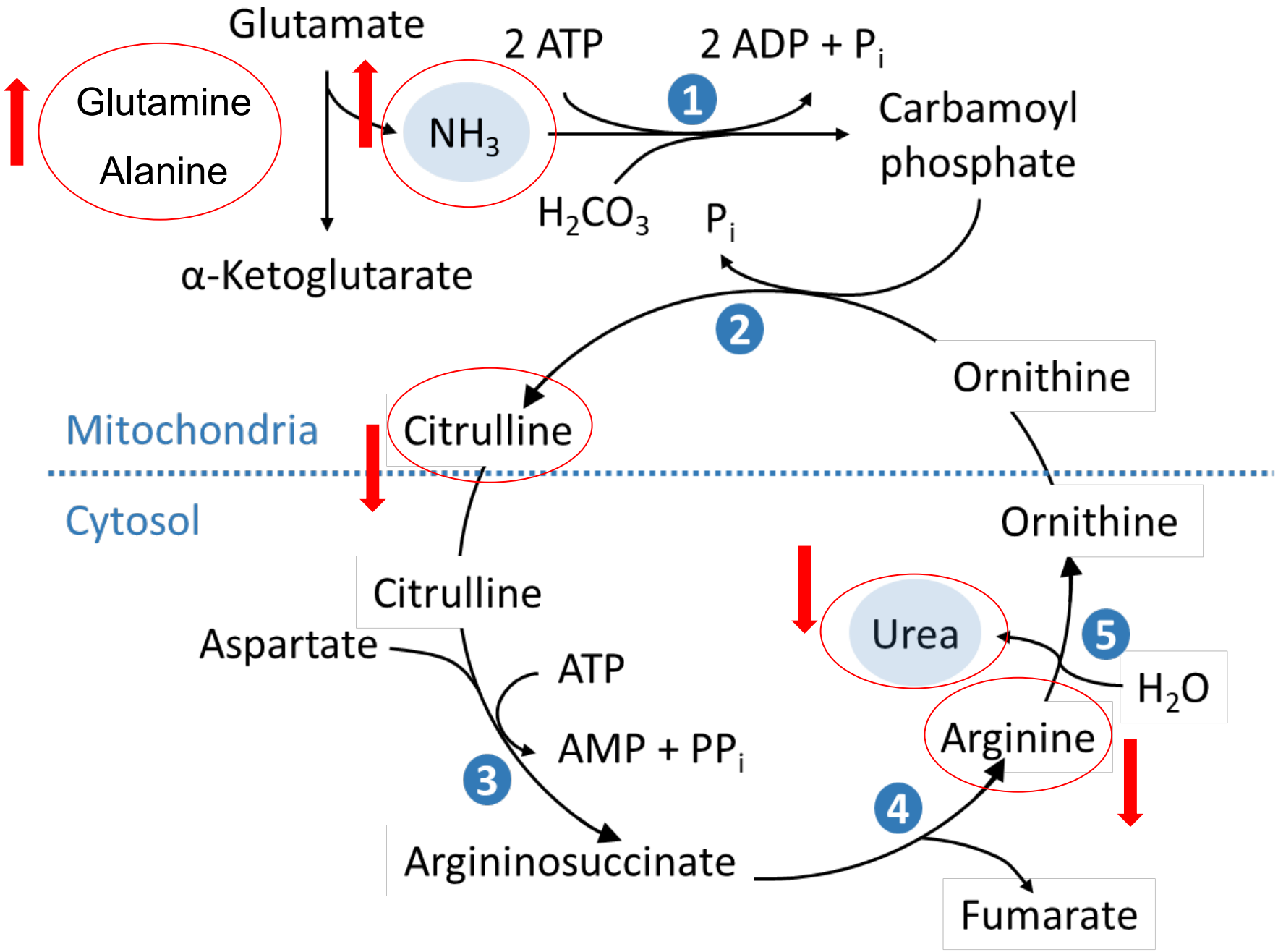
Hyperammonemia

- Liver damage
- Genetic disorders
  - N-acetylglutamate synthase deficiency
  - Carbamoyl phosphate synthetase I deficiency
  - Ornithine transcarbamoylase (OTC) deficiency
  - Ornithine translocase deficiency
  - Argininosuccinate synthetase deficiency (citrullinemia)
  - Argininosuccinate lyase deficiency (argininosuccinic aciduria)
  - Arginase deficiency (hyperargininemia)

Case report

A white male in his early 30s went to the emergency room after a four-wheeler accident. On the night following the orthopedic surgical procedure, he became confused and combative. He had tachycardia, hypertension and hyperthermia. The patient exhibited generalized tonic-clonic seizures. His medical history included chronic hypertension and persistent anemia with low hemoglobin levels. Discussions with the patient's family revealed that he was an "autoselective" vegetarian; he systematically picked out and rejected the meat in any dish served to him. He also had a history of nonspecific psychiatric problems.

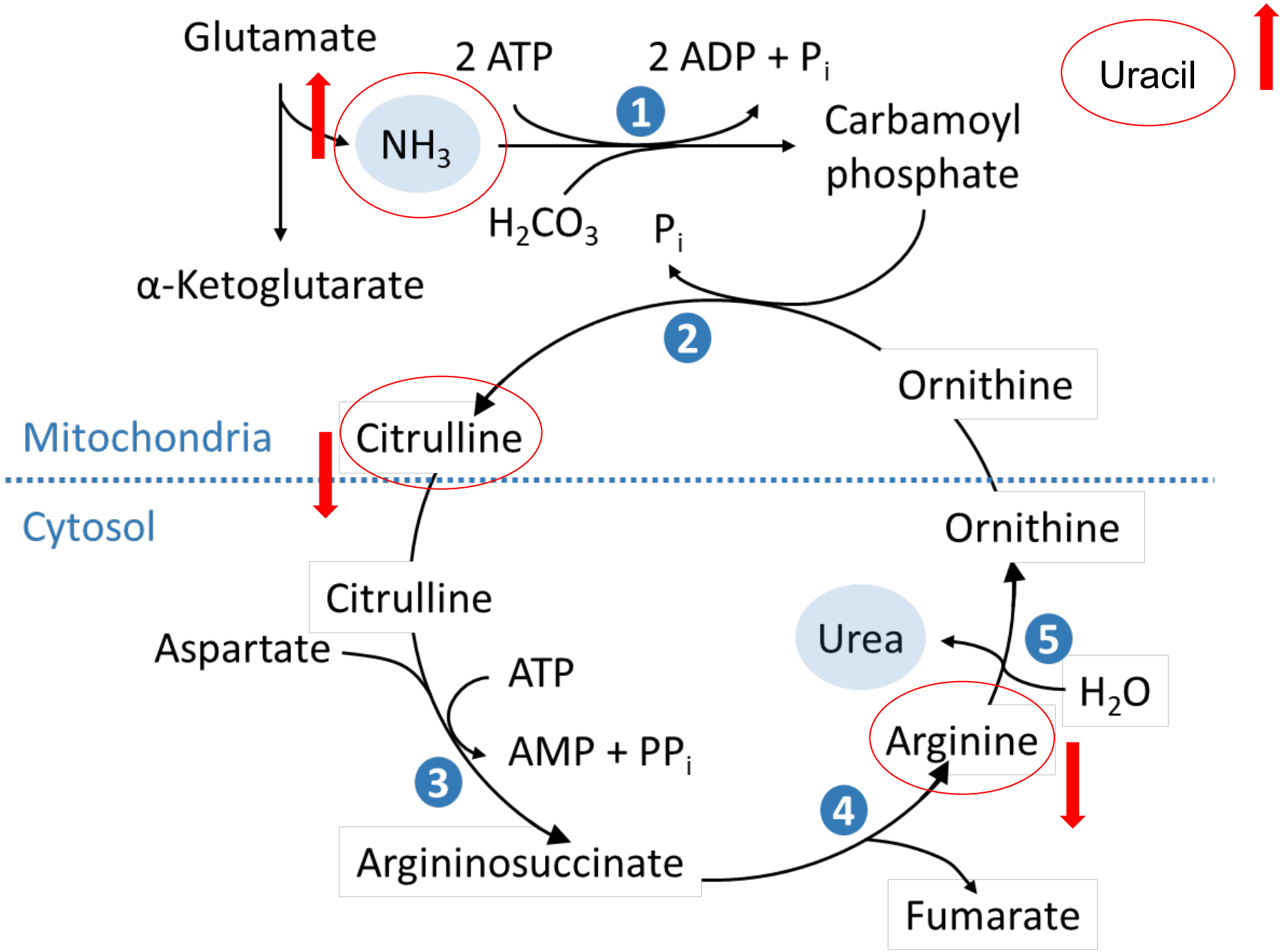
Laboratory tests revealed a low blood urea nitrogen and an ammonia level of 553 µmol/L. A plasma amino acid profile showed very high glutamine and high alanine levels, as well as very low citrulline and low arginine concentrations.


Glutamate
2 ATP
2 ADP + P_i
Glutamine
Alanine
NH_3
1
Carbamoyl phosphate
H_2CO_3
P_i
α-Ketoglutarate
2
Ornithine
Mitochondria
Citrulline
Cytosol
Ornithine
Citrulline
Urea
5
Aspartate
ATP
H_2O
Arginine
AMP + PP_i
3
4
Argininosuccinate
Fumarate

Case report

A 58-year-old female presented to her local hospital with recurrent episodes of "asthmatic bronchitis". Laboratory and diagnostic testing revealed an ammonia level of 280 μmol/L; cerebral edema on CT examination; very low citrulline and arginine; and an elevation of uracil in urine. She was on a regular diet, although the patient revealed that she tended to avoid large protein intakes.

Based on family history elicited from the patient's husband, the patient was diagnosed as a symptomatic OTC carrier. She had had four male infants who died from OTC deficiency. She has two adult daughters who are carriers of an OTC deficiency. One (a normal college graduate) lost a son. The other has mild intellectual disability and has suffered several mild hyperammonemic episodes in the past.


Glutamate
2 ATP
2 ADP + P_i
Uracil
NH_3
1
Carbamoyl phosphate
H_2CO_3
P_i
α-Ketoglutarate
2
Ornithine
Mitochondria
Citrulline
Cytosol
Ornithine
Citrulline
Urea
5
Aspartate
ATP
H_2O
AMP + PP_i
Arginine
3
4
Argininosuccinate
Fumarate

Clinical management of hyperammonemia

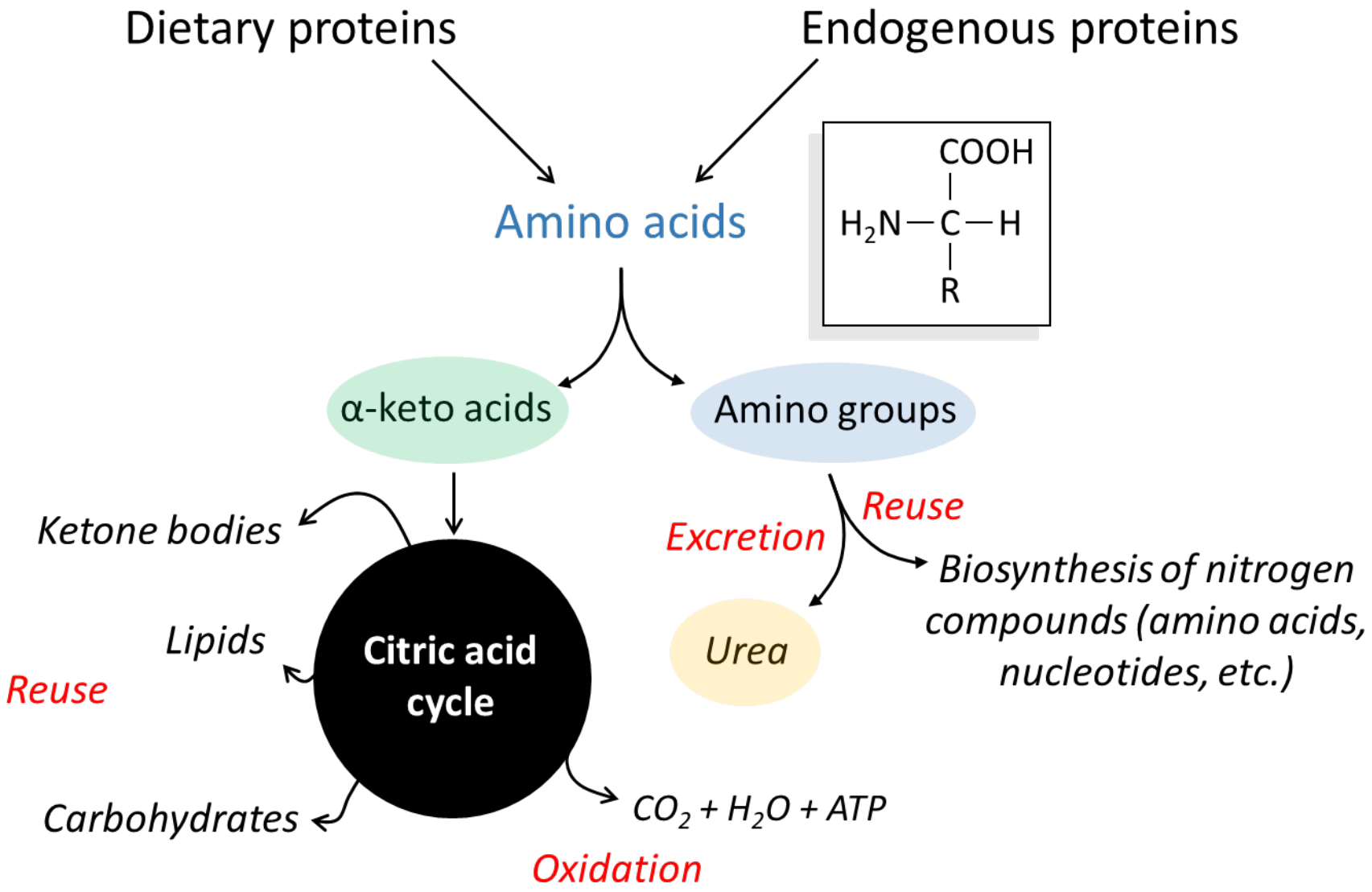
- Dietary protein restriction
- Hemodialysis
- Nitrogen scavengers
    - Sodium benzoate and sodium phenylbutyrate
- Liver transplantation

Nonessential and essential amino acids

TABLE 18-1	Nonessential and Essential Amino Acids for Humans and the Albino Rat	
Nonessential	**Conditionally essential[a]**	**Essential**
Alanine	Arginine	Histidine
Asparagine	Cysteine	Isoleucine
Aspartate	Glutamine	Leucine
Glutamate	Glycine	Lysine
Serine	Proline	Methionine
	Tyrosine	Phenylalanine
		Threonine
		Tryptophan
		Valine

[a]Required to some degree in young, growing animals and/or sometimes during illness.

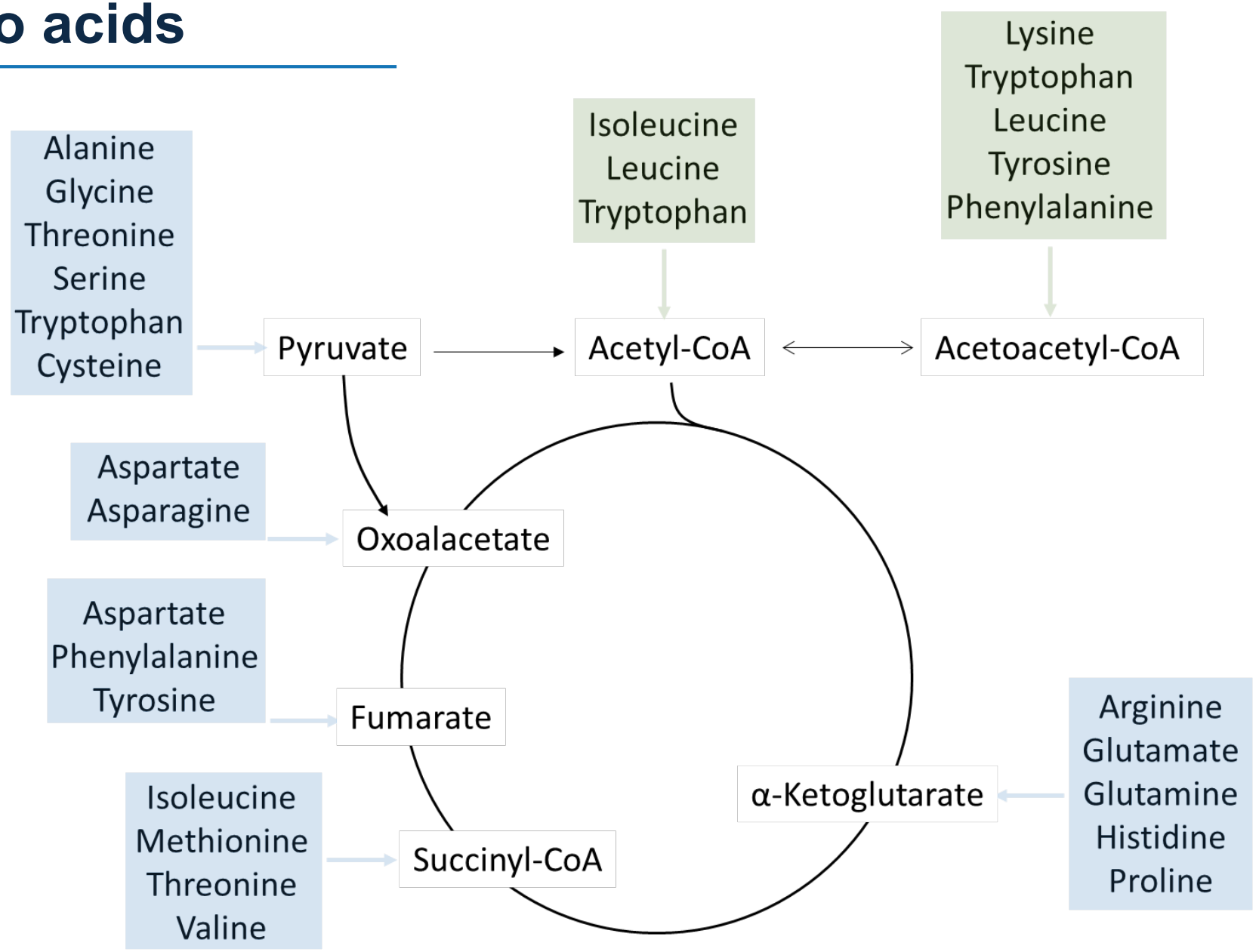
Outline of amino acid degradation



Catabolism of α-keto acids

Glucogenic amino acids

Ketogenic amino acids



Cofactors involved in amino acid catabolism

- Pyridoxal phosphate (PLP)
- Transfer of carbon-containing groups:
  - Biotin – carboxyl groups
  - Tetrahydrofolate (THF) – one-carbon units
  - S-adenosylmethionine (adoMet) – methyl groups
- Tetrahydrobiopterin (BH_4)

Pyridoxal phosphate (PLP)

- Derived from vitamin B_6
- Transaminations, isomerizations, modifications of amino acid side chains, decarboxylations, etc.

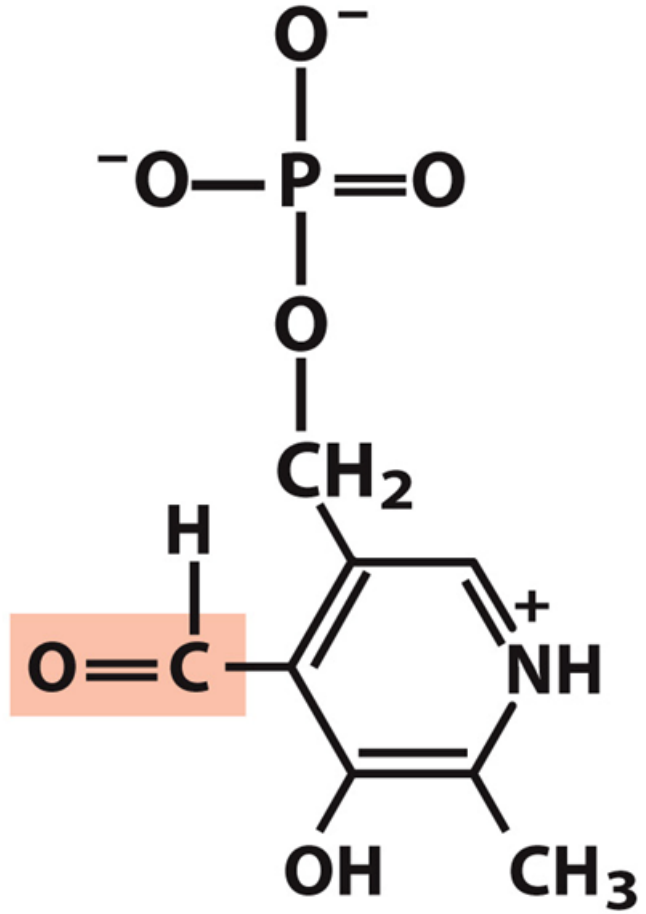


Figure 18-5a
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Biotin

- Vitamin B_7
- Carboxylations

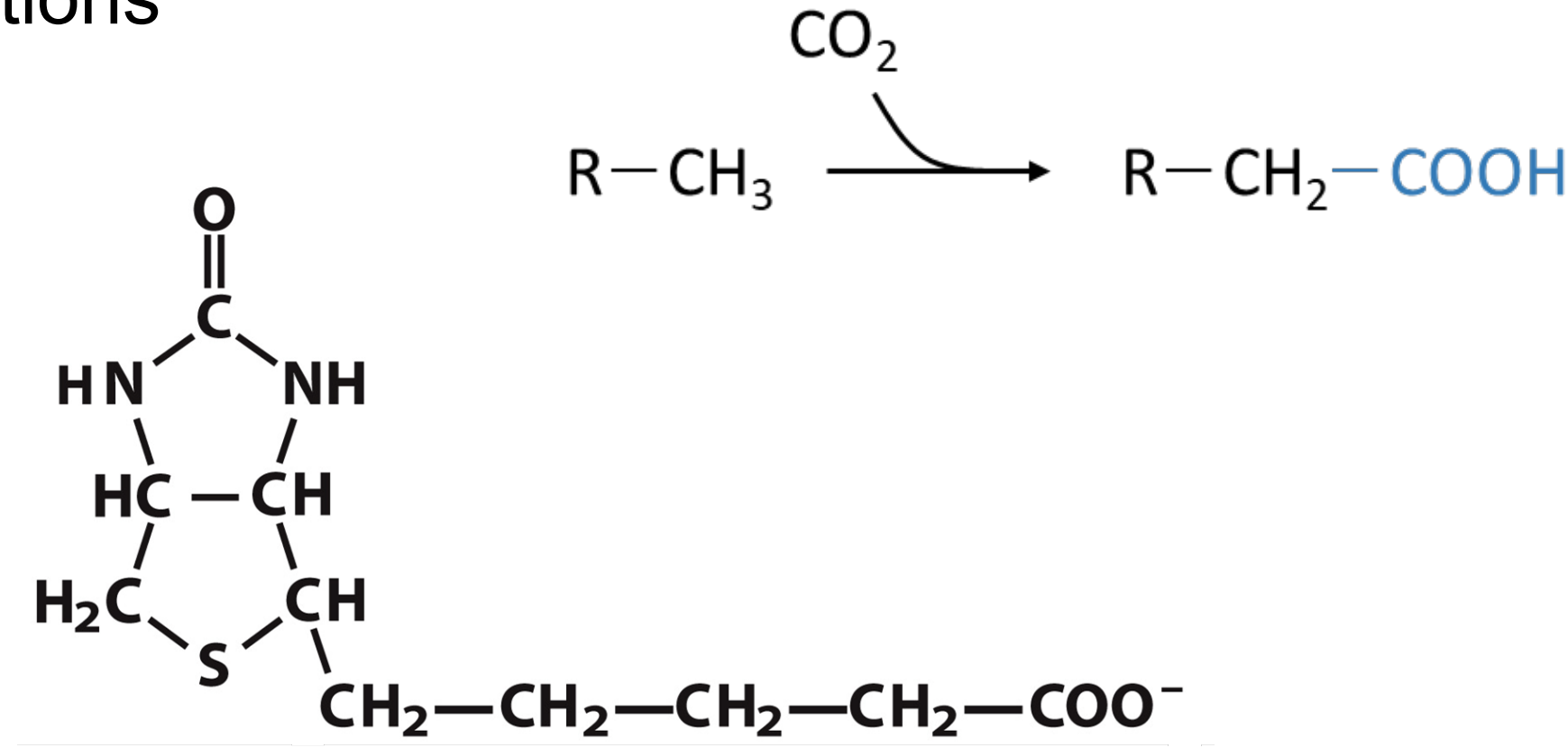


Figure 18-16 part 1
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Tetrahydrofolate (THF)

- Derived from folate (vitamin B_9)
- Transfer of one-carbon units at different oxidation states

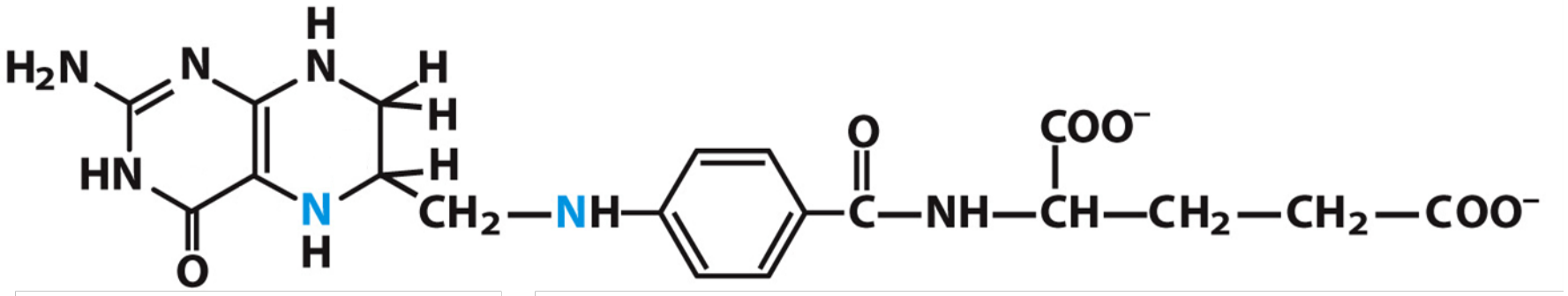
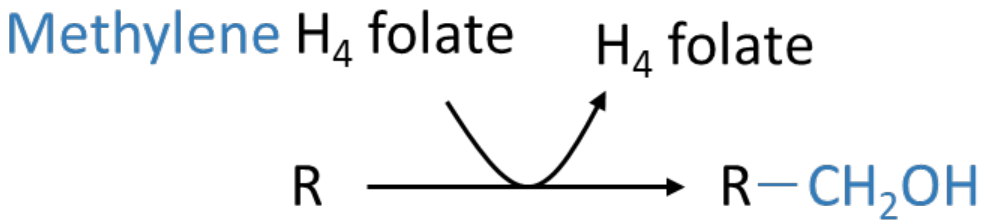
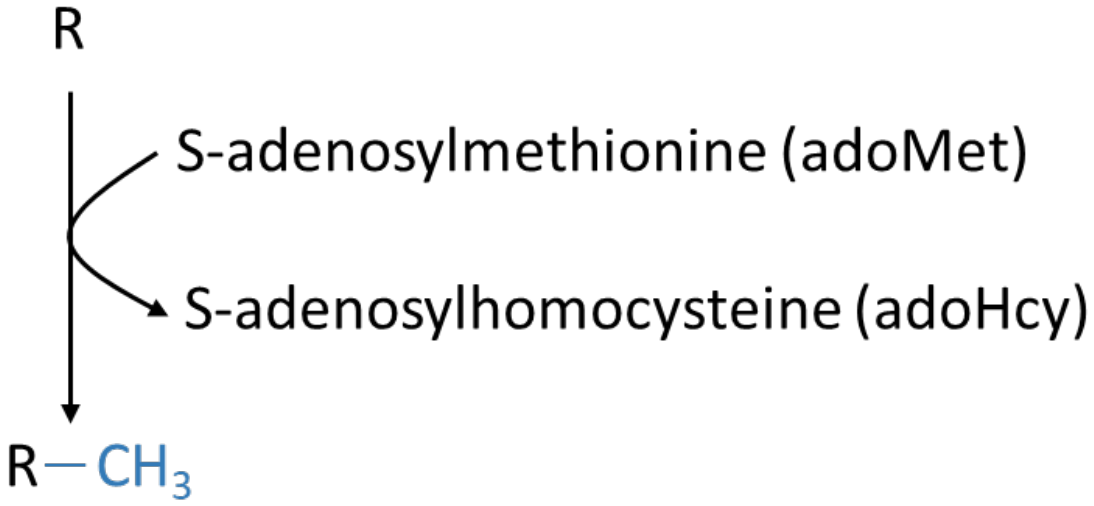


Figure 18-16 part 2
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S-adenosylmethionine

- SAM or AdoMet
- Methyl-group transfer reactions



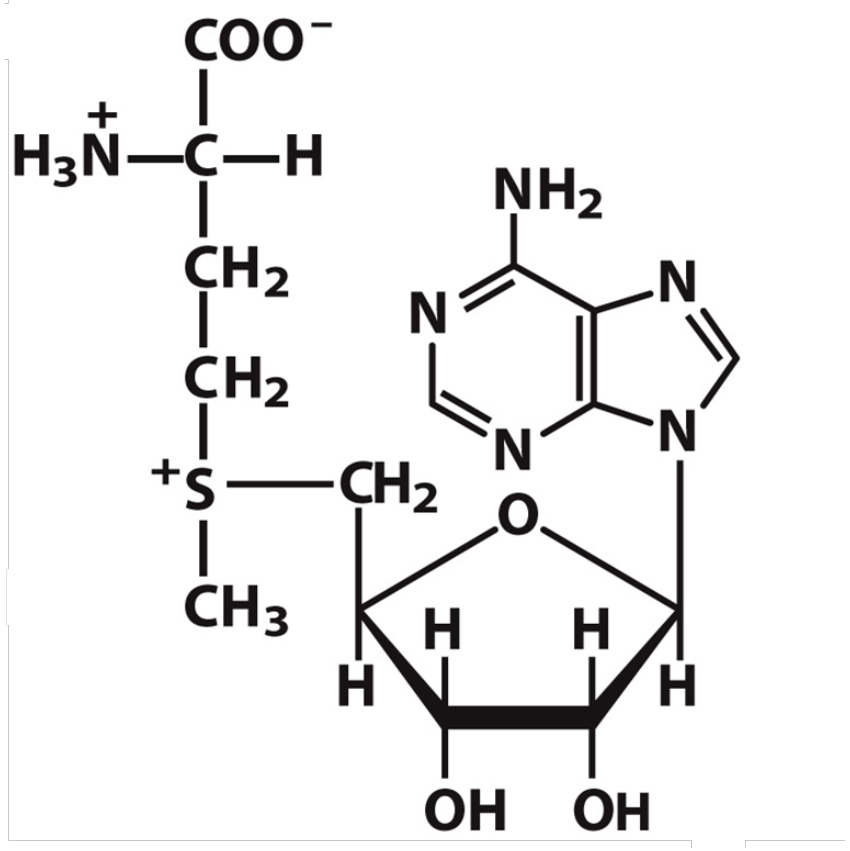


Figure 18-16 part 3
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Tetrahydrobiopterin (BH_4)

- Oxidation-reduction reactions

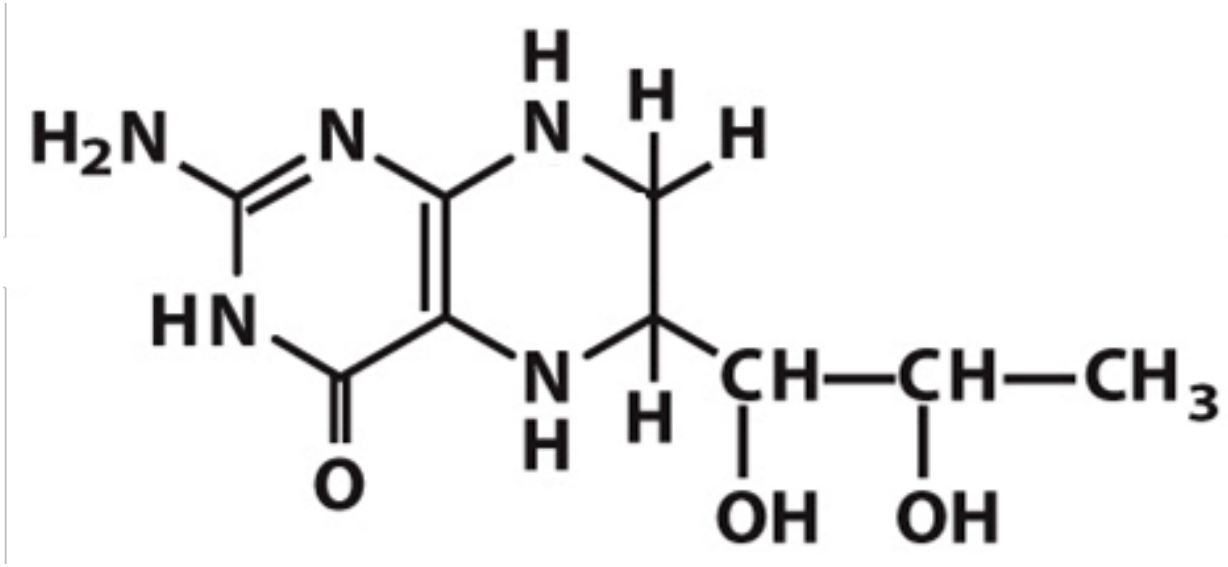


Figure 18-24
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Genetic disorders affecting amino acid catabolism

TABLE 18-2 Some Human Genetic Disorders Affecting Amino Acid Catabolism

Medical condition	Approximate incidence (per 100,000 births)	Defective process	Defective enzyme	Symptoms and effects
Albinism	<3	Melanin synthesis from tyrosine	Tyrosine 3- monooxygenase (tyrosinase)	Lack of pigmentation; white hair, pink skin
Alkaptonuria	<0.4	Tyrosine degradation	Homogentisate 1,2-dioxygenase	Dark pigment in urine; late-developing arthritis
Argininemia	<0.5	Urea synthesis	Arginase	Intellectual disability
Argininosuccinic acidemia	<1.5	Urea synthesis	Argininosuccinase	Vomiting; convulsions
Carbamoyl phosphate synthetase I deficiency	<0.5	Urea synthesis	Carbamoyl phosphate synthetase I	Lethargy; convulsions; early death
Homocystinuria	<0.5	Methionine degradation	Cystathionine β-synthase	Faulty bone development; intellectual disability
Maple syrup urine disease (branchedchain ketoaciduria)	<0.4	Isoleucine, leucine, and valine degradation	Branched-chain α-keto acid dehydrogenase complex	Vomiting; convulsions; intellectual disability; early death
Methylmalonic acidemia	<0.5	Conversion of propionyl-CoA to succinyl-CoA	Methylmalonyl-CoA mutase	Vomiting; convulsions; intellectual disability; early death
Phenylketonuria	<8	Conversion of phenylalanine to tyrosine	Phenylalanine hydroxylase	Neonatal vomiting; intellectual disability

Phenylketonuria

